

PMBOK / PMI | PMB-10

Work Breakdown Structure and WBS Dictionary

Hierarchical decomposition of the total scope into work packages

Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

Field	Detail
Document title	Work Breakdown Structure and WBS Dictionary
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Project / Programme	Helios Tranche 2 — Scale Build
Version	1.0
Status	Baselined
Date	18 Sep 2026
Author	D. Okonkwo, Programme Manager
Owner	Project manager with the delivery team
Approver	Sponsor
Classification	Internal
Retention	7 years from programme closure

Document control

Version history

Version	Date	Author	Summary of change
0.1	04 Jun 2026	D. Okonkwo, Programme Manager	Initial draft
0.2	21 Jul 2026	D. Okonkwo, Programme Manager	Updated following workstream lead review
1.0	18 Sep 2026	D. Okonkwo, Programme Manager	Baselined at the Tranche 2 funding gate

Reviewers

Name	Role	Review scope	Date reviewed
T. Nakamura, Chief Platform Architect	Chief Platform Architect	Technical content and solution alignment	08 Sep 2026
L. Haddad, CISO	CISO	Security, assurance and residency content	10 Sep 2026
C. Whitfield, Finance Business Partner	Finance Business Partner	Cost, funding and benefit figures	11 Sep 2026
M. Verity, Head of P3O	Head of P3O	Governance standards and completeness	15 Sep 2026

Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

Name	Role	Decision	Date
R. Al Mazrouei, Group CTO	Senior Responsible Owner / Group CTO	Approved	18 Sep 2026
H. Farrell, Director of Infrastructure	Director of Infrastructure	Approved	18 Sep 2026
L. Haddad, CISO	CISO	Approved subject to closure of ISS-003	18 Sep 2026

Related products

Reference	Product	Relationship
PMB-09	Project Scope Statement	The scope being decomposed
PMB-12	Activity and Milestone List	Activities decompose from work packages
PMB-16	Cost Baseline and Funding Requirements	Control accounts sit on the WBS

How to use this template

Aspect	Guidance
Purpose	Decomposes the project scope into a hierarchy of deliverables and work packages, and describes each element in the dictionary so that work can be estimated, scheduled, assigned and controlled.
Framework	PMBOK / PMI — Scope Management — Create WBS
Created when	During planning, after the scope statement
Reviewed / updated	Through change control only, once baselined
Owner	Project manager with the delivery team
Approver	Sponsor
Format options	Document, presentation, register entry, or tool record — the content matters, not the medium.
Tailoring	Decompose to the level at which you can reliably estimate and assign accountability, and no further. Over-decomposition creates control overhead without control.

This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.

Quality criteria

Use these to check the product is fit for purpose before submitting it for approval.

- Every element is a deliverable or outcome, not an activity
- The 100% rule holds — the children fully define the parent, with nothing added
- Work packages are small enough to estimate and assign to one accountable owner
- Every work package has a dictionary entry
- Project management effort itself appears in the WBS

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1. WBS structure

Present the hierarchy. A diagram or indented list both work — be consistent with the coding.

WBS code	Element	Level	Parent	Type	Owner
1.0	Programme management	1	—	Deliverable / Work package / Control account	D. Okonkwo, Programme Manager

2. WBS dictionary

One entry per work package. This is what turns a diagram into a control tool.

WBS code	Work package	Description of work	Acceptance criteria	Owner	Effort	Predecessors
1.0	Programme management	Planning, control, reporting, assurance and governance across all tranches	Accepted by the workstream lead against the WBS dictionary entry	D. Okonkwo, Programme Manager	4,200 h	Design freeze gate complete
2.0	Facilities and power	Power, cooling, structural and environmental works in DXB-1 halls A and B	Accepted by the workstream lead against the WBS dictionary entry	A. Kowalski, Data Centre Lead	3,100 h	Design freeze gate complete
2.1	Utility uplift to 6 MVA	Application, switchgear, energisation and acceptance of the additional supply	Accepted by the workstream lead against the WBS dictionary entry	A. Kowalski, Data Centre Lead	640 h	Design freeze gate complete
2.2	Liquid cooling distribution	CDUs, secondary loop, rear-door heat exchangers and leak detection	Accepted by the workstream lead against the WBS dictionary entry	A. Kowalski, Data Centre Lead	980 h	Design freeze gate complete
2.3	Structural strengthening	Floor loading assessment and strengthening for 1,400 kg racks	Accepted by the workstream lead against the WBS dictionary entry	A. Kowalski, Data Centre Lead	520 h	Design freeze gate complete
3.0	Compute and storage	Accelerator systems and the parallel filesystem tier	Accepted by the workstream lead against the WBS dictionary entry	T. Nakamura, Chief Platform Architect	2,600 h	Design freeze gate complete
3.1	Accelerator install and burn-in	Receipt, firmware verification, rack, cable, burn-in and admission to production	Accepted by the workstream lead against the WBS dictionary entry	S. Bhatt, Project Manager	1,480 h	Design freeze gate complete

WBS code	Work package	Description of work	Acceptance criteria	Owner	Effort	Predecessors
3.2	Parallel filesystem build	Storage hardware, filesystem configuration and throughput acceptance	Accepted by the workstream lead against the WBS dictionary entry	I. Petrova, Data Platform Lead	720 h	Design freeze gate complete
4.0	Network fabric	400G leaf-spine fabric, optics, cabling and performance tuning	Accepted by the workstream lead against the WBS dictionary entry	J. Ferreira, Network Lead	1,900 h	Design freeze gate complete
4.1	Fabric build and cabling	Leaf and spine installation, structured cabling and link validation	Accepted by the workstream lead against the WBS dictionary entry	J. Ferreira, Network Lead	1,100 h	Design freeze gate complete

3. Control accounts

The points at which scope, budget and schedule are integrated and performance is measured.

Control account	WBS elements included	Budget	Owner	Measurement method
1.0 Programme management	1.0 Programme management	AED 5,900,000	D. Okonkwo, Programme Manager	Milestone weighting at control account level
2.0 Facilities and power	2.0 Facilities and power	AED 18,200,000	A. Kowalski, Data Centre Lead	Milestone weighting at control account level
3.0 Compute and storage	3.0 Compute and storage	AED 28,400,000	T. Nakamura, Chief Platform Architect	Milestone weighting at control account level
4.0 Network fabric	4.0 Network fabric	AED 8,600,000	J. Ferreira, Network Lead	Milestone weighting at control account level
5.0 Platform and orchestration	5.0 Platform and orchestration	AED 2,400,000	P. Raghavan, Platform Engineering Lead	Milestone weighting at control account level
6.0 MLOps and developer experience	6.0 MLOps and developer experience	AED 900,000	N. Ozturk, ML Platform Lead	Milestone weighting at control account level
7.0 Security and compliance	7.0 Security and compliance	AED 1,400,000	K. Adeyemi, Security Architect	Milestone weighting at control account level

4. Assumptions used in decomposition

WBS code	Assumption	Impact if invalid
Handled through the Agile Release Train rather than the predictive workstreams	Vendor GPU allocation of 2,048 units for Tranche 2 will be honoured at the quoted delivery dates	Tranche 2 completion slips by at least one quarter and contract terms are renegotiated
Confirmed at the Tranche 1 end-of-tranche review, 18 Sep 2026	DXB-1 has 6 MVA of usable capacity available to the programme without a utility upgrade	Tranche 2 capped at 1,024 GPUs and the AUH-2 site is brought forward two quarters
Subject to closure of ISS-003 before multi-tenant go-live	Three business units will migrate committed workloads within six months of hall A handover	Utilisation benefit of AED 14.2m over five years is not realised in full
Applies across DXB-1 halls A and B for the Tranche 2 scale build	Training data classified as Restricted must remain within UAE borders at rest and in transit	Not applicable — this is a fixed constraint on design
Verified at hall A acceptance on 26 Feb 2027	Liquid cooling maintenance can be absorbed by the existing facilities contract without an uplift	Annual run cost increases by approximately AED 780,000

5. Excluded from the WBS

Anything a reader might expect to find but which is deliberately not present, and why.

Item	Why excluded	Where it sits instead
Programme management	Dependent on the Enterprise Identity Programme (DEP-003)	Escalated to the Portfolio Board where it exceeds programme tolerance
Facilities and power	Applies from hall A handover; hall B follows in Tranche 2 closure	Delegated to the Programme Manager within the agreed tolerance
Utility uplift to 6 MVA	Reviewed monthly at the Programme Board	Covers the 2,048-GPU estate and the parallel filesystem tier
Liquid cooling distribution	Subject to closure of ISS-003 before multi-tenant go-live	Maintained by the P3O as part of the assurance evidence library

